

Parasitic Finger: Coexistence with Artificial Organism

Akira Nakayasu
Tokyo Metropolitan University
Tokyo, Japan
nakayasu@tmu.ac.jp

Saki Sakaguchi
Tokyo Metropolitan University
Tokyo, Japan
s_saka@tmu.ac.jp

Mina Shibasaki
Tokyo Metropolitan University
Tokyo, Japan
shibasaki@tmu.ac.jp

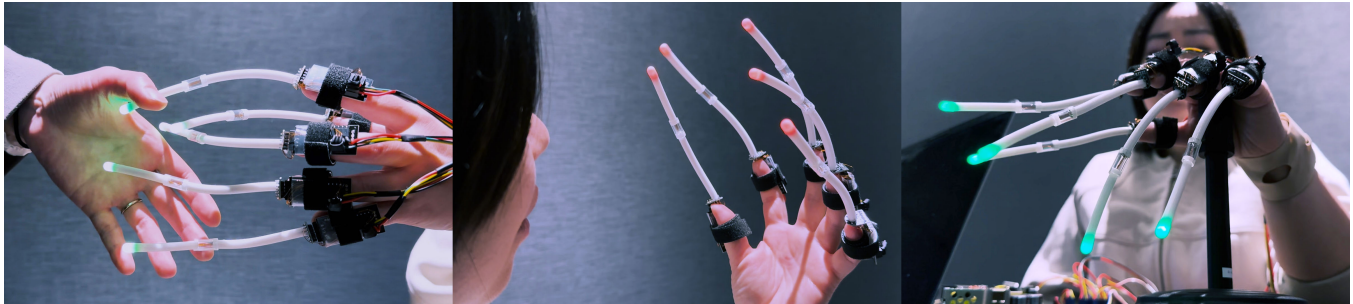


Figure 1: Snapshots of Parasitic Finger.

Abstract

The Parasitic Finger project explores how humans coexist with uncontrollable finger augmentation with an SMA actuator. What would happen if our fingers, which we use freely in daily life, could move in unrealistic ways, independent of our will? The finger actuator unit weighs just 5 grams and is lightweight, making it comfortable to wear on the fingertips. By regulating the heat of the SMA wires, the SMA actuator can bend in four directions. Unlike human finger joints, they move like tentacles, performing actions such as waving, touching objects, or grasping items. We designed the finger movements so that participants would feel a sense of biological agency or animacy.

CCS Concepts

• Human-centered computing → Human computer interaction (HCI); • Hardware → Emerging technologies.

Keywords

Human Augmentation, Cyborgs, Digital Cyborgs, Human-machine Integration, Soft Actuator, Shape-Memory Alloy

ACM Reference Format:

Akira Nakayasu, Saki Sakaguchi, and Mina Shibasaki. 2025. Parasitic Finger: Coexistence with Artificial Organism. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Emerging Technologies (SIGGRAPH Emerging Technologies '25)*, August 10–14, 2025, Vancouver, BC, Canada. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3721257.3734023>

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH Emerging Technologies '25, Vancouver, BC, Canada

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1551-8/25/08

<https://doi.org/10.1145/3721257.3734023>

1 Introduction

Body augmentation has been explored in the fields of research, art, and fashion [Nabeshima et al. 2019; Stelarc 1980; Wipprecht 2015]. Some of these enhance human physical capabilities, allowing users to control their extended bodies and express emotions. Some are beyond the user's control; they depend on their owner's body yet possess their own autonomy, moving independently like a parasite or mistletoe. This type of body augmentation, while part of the user's body, exists separately from the self, raising questions about the nature of embodiment and providing the sensation of coexisting with a different creature.

In this research, we focus on fingers as extensions of the body. Among body parts, fingers are the organs with the most precise control and sensitivity. They are frequently used in daily life, such as handling tools and operating equipment. Additionally, they convey information and express emotions through movements, including gestures and dance. What would happen if our fingers, which we use freely in daily life, could move in unrealistic ways, independent of our will? The user may try to use their extended fingers in some way, even if they aren't handling them well. They may sense another creature inside and feel an urge to engage with or care for it. This project investigates how humans coexist with uncontrollable finger augmentation. We delve into this concept by introducing the Parasitic Finger, a flexible finger extension that utilizes a shape-memory alloy (SMA) actuator.

2 Related works

Several studies have explored hand and finger extension. Examples of research include a method to increase the number of hands by attaching a robotic arm to the user's body [GOURMELEN et al. 2019], a method for operating a skeletal model of a hand smaller than the user's actual hand [Nishida et al. 2020], and a virtual hand that extends the length of the user's fingers using computer-generated images [Ogawa et al. 2016]. These augmentations are controllable, allowing users to manage the augmented hands and fingers, but our

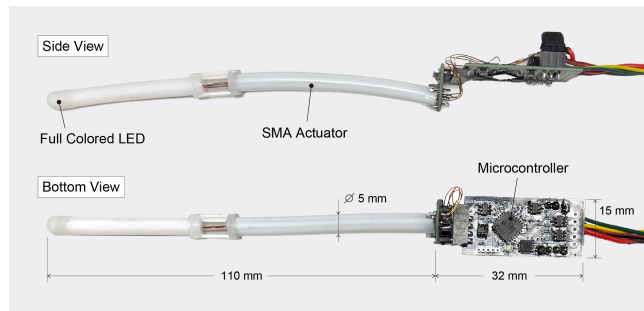


Figure 2: A finger actuator unit

research aims to develop uncontrollable augmentations. The study example for uncontrolled hand extension, [Minami et al. 2021], uses AR technology to project eyes and a mouth on the back of the user's hand, giving the user the sensation of coexisting with a living creature in their hand. In our research concept, instead of adding features of living beings like eyes and a mouth, we make the human finger uncontrollable by removing its inherent functionality. By allowing the extended finger to move independently, we aim to evoke a sense of life and coexistence with other organisms.

3 Parasitic Finger System

We have conducted several experiments and created interactive artworks inspired by the waving tentacles of sea anemones [Nakayasu 2020]. While previous work focused on art pieces utilizing SMA actuators, this project successfully developed a small, lightweight device that attaches to the fingertip. The Parasitic Finger system consists of a finger actuator unit and a control program developed using Cycling '74 Max on a PC. The control program communicates with the finger actuator unit via DMX 512, a serial communication protocol. It can control the actuator's behavior, including the light at the tip.

3.1 Finger Actuator Unit

The finger actuator unit includes an SMA actuator and a microcontroller board (Fig. 2). The SMA actuator is equipped with a silicon tube (5 mm in diameter and 110 mm long), four BioMetal fibers (a thin SMA wire developed by TOKI Corporation), a full-color LED chip, and a diffuser cap. These four BioMetal fibers drive the SMA actuator, enabling it to bend in four directions. The fibers contract when heated and return to their original length when cooled. The heating of each BioMetal fiber is controlled through pulse-width modulation (PWM) control currents. These PWM control currents are generated by a control circuit that comprises a peripheral interface controller (PIC), a field effect transistor, and various other electrical components. The microcontroller board measures 32 mm x 15 mm. The actuator unit weighs just 5 g, excluding the cable, and is so lightweight to wear on the fingertips.

3.2 Actuator Behavior

This system can control actuator behavior, including the bending direction in four directions, movement speed, and light color at the tip. Combining the movements of five actuators allows for

the expression of movements such as waving, touching objects, or grasping items. Unlike human finger joints, they can bend in four directions and show tentacle-like movements. The slow movement evokes an organic organism. Depending on the speed of the vibration, the creature's movement can express its alertness or, at times, its cuteness.

This project is still only an actuator-driven system. In the future, it will interact with physical information, such as myoelectric signals or brain waves, to enhance its affinity with the body. It will also connect to a network to convey environmental information, such as warnings of danger, and sometimes enjoy dancing to music. We could also upgrade it to a standalone, battery-powered model designed to fit the size of a hand.

4 Embodied Experience of the Parasitic Finger

The demo supports up to two participants at a time, who can experience the Parasitic Finger, a device that moves autonomously and cannot be voluntarily controlled. They engage in activities such as holding objects and shaking hands while wearing it. The Parasitic Finger moves randomly, requiring participants to adapt to its unpredictable behavior. Through this experience, we aim to provide a unique perspective on bodily extension and the challenges of coexisting with an autonomous, uncontrollable augmentation of the body.

Acknowledgments

HAYAO NAKAYAMA Foundation for Science & Technology and Culture supported this work.

References

- Guillaume GOURMELEN, Adrien Verhulst, Benjamin Navarro, Tomoya Sasaki, Ganesh Gowrishankar, and Masahiko Inami. 2019. Co-Limbs: An Intuitive Collaborative Control for Wearable Robotic Arms. In *SIGGRAPH Asia 2019 Emerging Technologies* (Brisbane, QLD, Australia) (SA '19). Association for Computing Machinery, New York, NY, USA, 9–10. doi:10.1145/3355049.3360526
- Asuka Minami, Hideyuki Takahashi, Yoshihiro Nakata, Hidenobu Sumioka, and Hiroshi Ishiguro. 2021. The Neighbor in My Left Hand: Development and Evaluation of an Integrative Agent System With Two Different Devices. *IEEE Access* 9 (2021), 98317–98326. doi:10.1109/ACCESS.2021.3095592
- Junichi Nabeshima, MHD Yamen Saraiji, and Kouta Minamizawa. 2019. Arque: artificial biomimicry-Inspired tail for extending innate body functions. In *ACM SIGGRAPH 2019 Emerging Technologies* (Los Angeles, California) (SIGGRAPH '19). Association for Computing Machinery, New York, NY, USA, Article 8, 2 pages. doi:10.1145/3305367.3327987
- Akira Nakayasu. 2020. Animated robotic sculptures: using SMA motion display to create lifelike movements. In *ACM SIGGRAPH 2020 Art Gallery* (Virtual Event, USA) (SIGGRAPH '20). Association for Computing Machinery, New York, NY, USA, 419–423. doi:10.1145/3386567.3388572
- Jun Nishida, Soichiro Matsuda, Hiroshi Matsui, Shan-Yuan Teng, Ziwei Liu, Kenji Suzuki, and Pedro Lopes. 2020. HandMorph: a Passive Exoskeleton that Miniaturizes Grasp. In *Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology* (Virtual Event, USA) (UIST '20). Association for Computing Machinery, New York, NY, USA, 565–578. doi:10.1145/3379337.3415875
- Nami Ogawa, Yuki Ban, Sho Sakurai, Takuji Narumi, Tomohiro Tanikawa, and Michitaka Hirose. 2016. Metamorphosis Hand: Dynamically Transforming Hands. In *Proceedings of the 7th Augmented Human International Conference 2016* (Geneva, Switzerland) (AH '16). Association for Computing Machinery, New York, NY, USA, Article 51, 2 pages. doi:10.1145/2875194.2875246
- Stelarc. 1980. *THE THIRD HAND*. Retrieved February 10, 2025 from http://stelarc.org/_php#page/2
- Anouk Wipprecht. 2015. *Robotic Spider Dress*. Retrieved February 10, 2025 from <https://vimeo.com/114828162>